

EcoPackAI - Complete Material Database

Generated on: April 06, 2026

CO2_Emission_Score	Recyclability_Percentage	Data_Source	Weight_kg	CO2_Impact_Index	Cost_Efficiency_Index	Material_Suitability_Score	Industry_encoded	Material_Type_encoded
0.0	0.896551724137931	Carbon Catalogue	0.1825667804780076	0.8455172413793104	0.8174332195219924	0.775436862408322	1	0
0.7428571428571429	0.1379310344827586	OpenLCA Sample	0.4675637678248644	0.3850246305418719	0.5324362321751356	0.5417407218692896	3	5
0.8428571428571427	0.2413793103448275	Sustainability Dataset	0.1656959228760795	0.1817296113847838	0.8343040771239205	0.3669830676910897	6	4
0.5857142857142856	0.2241379310344827	Sustainability Dataset	0.1231170917854991	0.4794745484400657	0.8768829082145009	0.6948546918403765	5	5
0.8428571428571427	0.2758620689655171	Carbon Catalogue	0.4119301064470777	0.4154460864805693	0.5880698935529223	0.5865994026581044	4	5
0.5857142857142856	0.293103448275862	FootprintData	0.7702349869451696	0.5380186097427476	0.2297650130548304	0.5561369478135483	3	5
0.5142857142857142	0.6551724137931034	Carbon Catalogue	0.2982526611769432	0.429184455391352	0.7017473388230568	0.4261979838034579	0	5
0.7285714285714286	0.2586206896551724	Sustainability Dataset	0.1936131753364129	0.4522386425834702	0.8063868246635871	0.6708115044324643	4	4
0.1	0.2586206896551724	Sustainability Dataset	0.0859610363526812	0.5595402298850575	0.9140389636473188	0.6540277810482187	0	0
0.9714285714285716	0.8793103448275861	OpenLCA Sample	0.318738702550713	0.4004050355774493	0.6812612974492871	0.4525404034657659	2	4
0.3285714285714286	0.1724137931034482	Carbon Catalogue	0.1612773649327174	0.5079474548440066	0.8387226350672826	0.6587957724577874	8	2
0.5285714285714285	0.6896551724137931	FootprintData	0.5768226551516369	0.680361247947455	0.4231773448483631	0.663097702633491	1	1
0.657142857142857	0.3275862068965516	OpenLCA Sample	0.8949588270737094	0.2634811165845649	0.1050411729262906	0.1729047985117131	2	2
0.4142857142857143	0.3103448275862069	Carbon Catalogue	0.0636674030929905	0.6097974822112754	0.9363325969070094	0.8048187719566131	2	5
0.4999999999999999	0.603448275862069	FootprintData	0.5019080136573609	0.6611494252873563	0.4980919863426391	0.6778873660177342	6	1
0.7571428571428571	0.1724137931034482	OpenLCA Sample	0.0622614982928298	0.4628680897646415	0.93773850170717	0.7584687864180075	1	0
0.657142857142857	0.396551724137931	FootprintData	0.846153846153846	0.2642474001094691	0.153846153846154	0.1678528061976338	2	2
0.1285714285714286	0.8448275862068966	Sustainability Dataset	0.0435830488049809	0.6120853858784894	0.9564169511950192	0.5677592397099016	0	5
0.6142857142857143	0.8275862068965516	Sustainability Dataset	0.623016670014059	0.4533223864258347	0.376983329985941	0.3384239535661162	4	1
0.2142857142857143	0.6896551724137931	Carbon Catalogue	0.1317533641293432	0.7406787082649152	0.8682466358706568	0.7807454740671631	4	5

O2_Emission_Score	Recyclability_Percentage	Data_Source	Weight_kg	CO2_Impact_Index	Cost_Efficiency_Index	Material_Suitability_Score	Industry_encoded	Material_Type_encoded
0.7857142857142856	0.2758620689655171	Carbon Catalogue	0.3711588672424181	0.3322714833059661	0.6288411327575819	0.4735609331496611	5	5
0.7	0.0344827586206896	Sustainability Dataset	0.9897569793131148	0.1470498084291188	0.0102430206868851	0.093892829577713	6	2
0.3571428571428572	0.8103448275862069	FootprintData	0.7724442659168507	0.8088451012588944	0.2275557340831493	0.6838047607285025	0	4
0.7714285714285714	0.5	Carbon Catalogue	0.1462140992167102	0.4339682539682541	0.8537859007832899	0.6017230718222886	5	2
0.5285714285714285	0.1379310344827586	FootprintData	0.0377585860614581	0.3497865353037766	0.962241413938542	0.5605870383030732	0	0
0.9142857142857144	0.0	FootprintData	0.9945772243422372	0.3485714285714286	0.0054227756577627	0.4290554041259003	1	5
0.0	0.7586206896551724	OpenLCA Sample	0.6469170516167904	0.6306513409961686	0.3530829483832096	0.3981854209134303	7	0
0.5285714285714285	0.189655172413793	Carbon Catalogue	0.6404900582446275	0.3803612479474549	0.3595099417553725	0.4039974817055937	7	4
0.3142857142857143	0.9310344827586206	OpenLCA Sample	0.6969270937939345	0.8678051450465244	0.3030729062060655	0.7340439298804295	4	5
0.4428571428571429	0.5517241379310345	FootprintData	0.5926892950391645	0.4496223316912972	0.4073107049608355	0.3740421441647696	3	5
0.9142857142857144	0.5689655172413792	FootprintData	0.1964249849367343	0.4626710454296661	0.8035750150632657	0.6461409226908461	4	2
0.9285714285714286	0.4482758620689655	FootprintData	0.7304679654549106	0.2754570333880679	0.2695320345450894	0.283042423718754	1	5
0.8285714285714285	0.7758620689655171	OpenLCA Sample	0.5884715806386824	0.3913191023535851	0.4115284193613176	0.3479861667498293	8	0
0.657142857142857	0.103448275862069	OpenLCA Sample	0.0574412532637075	0.1532129173508484	0.9425587467362924	0.3480527909612271	3	1
0.9714285714285716	0.603448275862069	Carbon Catalogue	0.6748343040771239	0.3662287903667214	0.3251656959228761	0.3840412249235514	3	2
0.9571428571428572	0.7413793103448276	Carbon Catalogue	0.0696927093793934	0.5769677066228791	0.9303072906206066	0.7938792698353336	0	5
0.8714285714285714	0.3275862068965516	OpenLCA Sample	0.1843743723639285	0.2676081007115489	0.8156256276360715	0.4557309285754411	3	5
0.5285714285714285	0.7413793103448276	FootprintData	0.133360112472384	0.5109359605911331	0.866639887527616	0.560366350494738	0	5
0.8571428571428572	0.7413793103448276	FootprintData	0.4717814822253464	0.5614121510673235	0.5282185177746536	0.6230304157593254	0	2
0.7	0.3103448275862069	FootprintData	0.1988351074512954	0.3545593869731801	0.8011648925487046	0.5181732225538834	1	4
0.6428571428571428	0.2241379310344827	Sustainability Dataset	0.8660373568989757	0.3093158182813355	0.1339626431010243	0.2679151202428415	5	0
0.2857142857142857	0.6379310344827585	Sustainability Dataset	0.5474994978911427	0.4818500273672687	0.4525005021088573	0.3564901615795647	7	5
0.3	0.6206896551724137	FootprintData	0.3008636272343844	0.5868965517241379	0.6991363727656157	0.5764995325193398	2	3
0.2142857142857143	0.8448275862068966	Carbon Catalogue	0.2986543482627033	0.7479584017515052	0.7013456517372967	0.6935870562217911	2	5

Comprehensive Environmental Impact Analysis: Q3 2023								
Product Line Overview			Material Composition			Manufacturing & Logistics		
Product ID	Product Name	Category	Material A (%)	Material B (%)	Material C (%)	Production Volume	Shipping Weight (kg)	Lead Time (days)
CO2 Emission Score	Recyclability Percentage	Data Source	Weight_kg	CO2_Impact_Index	Cost_Efficiency_Index	Material_Suitability_Score	Industry_encoded	Material_Type_code
0.2285714285714286	0.982758620689655	OpenLCA Sample	0.2325768226551516	0.8113957307060756	0.7674231773448484	0.7587852454858849	7	4
0.7	0.293103448275862	OpenLCA Sample	0.385418758786905	0.5265900383141763	0.6145812412130951	0.6910103876895991	6	0
0.657142857142857	0.6206896551724137	FootprintData	0.5872665193814018	0.3878489326765189	0.4127334806185982	0.338959617256187	8	2
0.7857142857142856	0.8793103448275861	OpenLCA Sample	0.087567784695722	0.6178653530377669	0.912432215304278	0.7488758058063902	4	2
0.1142857142857143	0.0517241379310344	Sustainability Dataset	0.0755171721229162	0.5835905856595512	0.9244828278770838	0.7547810826269457	8	5
0.9142857142857144	0.3620689655172413	Sustainability Dataset	0.2337818839124322	0.2425944170771757	0.7662181160875678	0.4109032016571406	8	0
0.5285714285714285	0.3620689655172413	FootprintData	0.7535649728861217	0.5444991789819377	0.2464350271138783	0.5317301797269386	7	1
0.5428571428571429	0.293103448275862	OpenLCA Sample	0.2335810403695521	0.4278598795840176	0.7664189596304479	0.5610696397227415	8	5
0.2142857142857143	0.8448275862068966	OpenLCA Sample	0.5021088572002409	0.7212917350848386	0.4978911427997591	0.5978840368738632	2	0
0.0714285714285714	0.2586206896551724	OpenLCA Sample	0.379393452500502	0.449064039408867	0.620606547499498	0.4138075800133962	2	5
0.7571428571428571	0.982758620689655	Sustainability Dataset	0.3388230568387226	0.5640941434044883	0.6611769431612774	0.5639907403101785	3	5
0.7285714285714286	0.8103448275862069	OpenLCA Sample	0.0712994577224342	0.5739244663382594	0.9287005422775658	0.7001799492185735	3	5
0.5142857142857142	0.4655172413793103	OpenLCA Sample	0.3340028118096002	0.392632731253421	0.6659971881903998	0.4248522489584883	4	2
0.8571428571428572	0.2068965517241379	OpenLCA Sample	0.1773448483631251	0.3299178981937603	0.822655151636875	0.5707637047685666	8	5
0.6714285714285715	0.9482758620689656	OpenLCA Sample	0.6370757180156658	0.5145046524356869	0.3629242819843342	0.3946791455695751	3	5
0.3	0.1551724137931034	FootprintData	0.1132757581843744	0.3517241379310345	0.8867242418156256	0.4667069277171015	4	4
0.8428571428571427	0.1724137931034482	OpenLCA Sample	0.6818638280779272	0.1898522167487685	0.3181361719220728	0.2433817382761292	1	1
0.8	0.8620689655172413	FootprintData	0.6013255673830086	0.362911877394636	0.3986744326169914	0.2727670807429518	2	0
0.0428571428571428	0.482758620689655	OpenLCA Sample	0.3623217513556938	0.5821893814997262	0.6376782486443062	0.5161792271931823	5	2
0.7428571428571429	0.3620689655172413	OpenLCA Sample	0.3826069491865836	0.4730706075533662	0.6173930508134164	0.6144461582653714	3	0
0.2571428571428571	0.4482758620689655	Carbon Catalogue	0.2580839526009239	0.4637110016420361	0.7419160473990761	0.4680592148765373	6	0
0.2857142857142857	0.482758620689655	Carbon Catalogue	0.974894557139988	0.656792556102901	0.025105442860012	0.5022486552991641	0	5
0.3428571428571429	0.3275862068965516	Sustainability Dataset	0.5693914440650732	0.5193541324575808	0.4306085559349268	0.5089242197635104	1	5
0.9285714285714286	0.3448275862068966	FootprintData	0.474994978911428	0.2320853858784893	0.5250050210885719	0.3343356606779673	0	2

Comprehensive Environmental Impact Analysis Report - Q3 2023								
Product Identification			Material Properties			Performance Metrics		
CO2_Emission_Score	Recyclability_Percentage	Data_Source	Weight_kg	CO2_Impact_Index	Cost_Efficiency_Index	Material_Suitability_Score	Industry_encoded	Material_Type_code
0.3428571428571429	0.7241379310344827	FootprintData	0.2904197630046194	0.6070935960591133	0.7095802369953806	0.5877115095222596	1	1
0.0714285714285714	0.7241379310344827	OpenLCA Sample	0.262502510544286	0.6309031198686371	0.737497489455714	0.5456104947841691	0	5
0.1857142857142857	0.189655172413793	OpenLCA Sample	0.2841936131753364	0.4368691844553913	0.7158063868246636	0.4814895898295556	2	5
0.5857142857142856	0.603448275862069	Sustainability Dataset	0.5515163687487447	0.4370224411603723	0.4484836312512553	0.3973540658395255	1	3
0.2285714285714286	0.9655172413793104	FootprintData	0.6212090781281381	0.8500930487137385	0.3787909218718619	0.697674496047054	5	3
0.1714285714285714	0.9137931034482758	Sustainability Dataset	0.4099216710182767	0.5896770662287903	0.5900783289817233	0.4208943251860331	8	3
0.7571428571428571	0.8793103448275861	Sustainability Dataset	0.7615987146013254	0.5073891625615764	0.2384012853986746	0.394476050644233	7	5
0.1571428571428571	0.0344827586206896	Sustainability Dataset	0.0492066680056236	0.3502244116037219	0.9507933319943764	0.4773277642398017	7	4
0.9428571428571428	0.603448275862069	OpenLCA Sample	0.1893954609359309	0.4735303776683088	0.810604539064069	0.6605935127865443	4	2
0.5428571428571429	0.1379310344827586	Sustainability Dataset	0.1096605744125326	0.5228024083196496	0.8903394255874674	0.7682227910041	5	2
0.4857142857142857	0.4482758620689655	OpenLCA Sample	0.2153042779674633	0.5297427476737822	0.7846957220325367	0.6353058156792739	8	0
0.2142857142857143	0.1551724137931034	Carbon Catalogue	0.0650733078931512	0.3580733442802408	0.9349266921068488	0.4637073453441509	1	5
0.9714285714285716	0.8620689655172413	Carbon Catalogue	0.2715404699738903	0.5902134646962233	0.7284595300261096	0.7186232448863222	7	2
0.2857142857142857	0.2586206896551724	FootprintData	0.3912432215304277	0.4531910235358511	0.6087567784695722	0.4799034429552121	5	2
0.7571428571428571	0.2413793103448275	OpenLCA Sample	0.614581241213095	0.2414121510673235	0.385418758786905	0.2841904880630009	8	5
0.8571428571428572	0.8620689655172413	Sustainability Dataset	0.002209278971681	0.4105309250136837	0.9977907210283188	0.5315495863139692	1	5
0.7285714285714286	0.7241379310344827	OpenLCA Sample	0.4898574010845551	0.5140777230432402	0.5101425989154449	0.5226738688919295	2	2
0.6285714285714286	0.2413793103448275	FootprintData	0.0036151837718417	0.2487137383689108	0.9963848162281582	0.4384009402160117	7	5
0.7714285714285714	1.0	FootprintData	0.129142398071902	0.5784126984126985	0.870857601928098	0.6446223599435088	5	4
0.5428571428571429	0.4482758620689655	FootprintData	0.8702550712994577	0.3240284619594964	0.1297449287005423	0.1885348633939613	8	5
0.1	0.1724137931034482	OpenLCA Sample	0.1871861819642498	0.5219157088122605	0.8128138180357501	0.6006104289356292	7	4
0.2142857142857143	0.3448275862068966	FootprintData	0.596103635268126	0.6568472906403942	0.403896364731874	0.6359078256757199	5	5
0.6285714285714286	0.1551724137931034	Sustainability Dataset	0.93793934525005	0.322200328407225	0.0620606547499499	0.279498327787875	3	0
0.4571428571428571	0.5172413793103448	OpenLCA Sample	0.3950592488451496	0.6066995073891627	0.6049407511548504	0.6521620283021202	3	0

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0.1857142857142857	0.5	Sustainability Dataset	0.4974894557139987	0.6736507936507937	0.5025105442860013	0.6322134807461179	7	3
0.7857142857142856	0.086206896551724	OpenLCA Sample	0.8702550712994577	0.4023864258347018	0.1297449287005423	0.4718780489440434	5	5
0.2571428571428571	0.2068965517241379	FootprintData	0.6420968065876682	0.4099178981937603	0.3579031934123318	0.3553381173012037	7	5
0.9714285714285716	0.2758620689655171	OpenLCA Sample	0.4868447479413537	0.1148111658456485	0.5131552520586463	0.2118710419558533	1	5
0.2714285714285714	0.4310344827586206	Sustainability Dataset	0.2753564972886122	0.5909797482211275	0.7246435027113878	0.6377849501018673	5	2
0.6285714285714286	0.9137931034482758	Carbon Catalogue	0.993773850170717	0.717296113847838	0.006226149829283	0.5487862904879202	3	2
0.2714285714285714	0.6206896551724137	Carbon Catalogue	0.4261899979915645	0.6541981390257252	0.5738100020084355	0.6178222562128207	0	5
0.7428571428571429	0.6206896551724137	Sustainability Dataset	0.222936332596907	0.3103886152162014	0.777063667403093	0.3732745463074084	2	5